Math 2650 – Quiz 05 Concept Practice

Choose the section matching the skill you want to practice. Answers are on the back.

A. Level Curves and Level Surfaces

(a) Let

$$f(x,y,z) = \int_0^x e^t\,dt + \int_1^y \frac{dt}{t} + \sqrt{z-1}.$$

Find the domain of f and the level surface through $(\ln 2, e, 5)$.

(b) For

$$f(x,y) = \begin{cases} \dfrac{x^2+y^2}{2x+y}, & 2x+y \neq 0, \\ 0, & 2x+y = 0, \end{cases}$$

sketch and label the level curves of heights -2 and 2.

(c) Let

$$f(x,y,z) = \frac{2x-y+z+1}{x+3y-2z+4}.$$

Find an equation of the level surface through $(1,0,2)$.

B. Gradient and Directional Change

(a) Find the direction of steepest descent at $(1,0)$ for

$$f(x,y) = e^y(x^2+xy) + (x-y)^3.$$

(b) For

$$f(x,y) = x^2 + xy + 2y^2,$$

find the unique vector $\mathbf{v} = \langle a, b\rangle$ at $(1,-1)$ such that f has zero change in the direction of $\mathbf{v}$,

$$|\mathbf{v}| = 2\sqrt{10}, \qquad a > 0.$$

C. Partial and Implicit Differentiation

(a) For

$$f(x,y) = \log_{x+1}(y^2+1),$$

find f_x and f_y.

(b) The equation

$$x^2 + 2y^2 + z^2 + xyz = 5$$

defines z implicitly near $(1,1,1)$. Find

$$\frac{\partial z}{\partial y}$$

at $(1,1,1)$.

D. Multivariable Chain Rule

(a) Suppose

$$F(x,y) = H\big(p(x,y), q(x,y)\big).$$

At $(1,-1)$,

$$p=-1, \quad q=2, \quad p_x=2, \quad p_y=-3, \quad q_x=1, \quad q_y=4,$$

and

$$H_p(-1,2) = 3, \qquad H_q(-1,2) = 2.$$

Find $F_x(1,-1)$ and $F_y(1,-1)$.

(b) Let

$$g(s,t) = f(s+t^2,\ st,\ s^2-t).$$

Given

$$\nabla f(5,2,-1) = \langle 3,-2,4\rangle,$$

find $g_t(1,2)$.

(c) Let

$$f(x,y,z) = xyz + e^{x^2+y^2}.$$

Using

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

express f_r, f_θ, and f_z in terms of r, θ, z.

Answer Key

Final answers only

A. Level Curves and Level Surfaces

(a)

$$\boxed{x \in \mathbb{R}, \qquad y > 0, \qquad z \geq 1}$$

and

$$\boxed{e^x + \ln y + \sqrt{z-1} = 5}.$$

(b) Height -2:

$$\boxed{(x+2)^2 + (y+1)^2 = 5}.$$

Height 2:

$$\boxed{(x-2)^2 + (y-1)^2 = 5}.$$

For both level curves,

$$\boxed{(0,0) \text{ is excluded}}.$$

(c)

$$\boxed{3x + 16y - 11z = -19}.$$

B. Gradient and Directional Change

(a)

$$\boxed{\frac{1}{\sqrt{26}}\langle -5, 1\rangle}.$$

(b)

$$\boxed{\mathbf{v} = \langle 6, 2\rangle}.$$

C. Partial and Implicit Differentiation

(a)

$$\boxed{f_x = -\frac{\ln(y^2+1)}{(x+1)[\ln(x+1)]^2}}, \qquad \boxed{f_y = \frac{2y}{(y^2+1)\ln(x+1)}}.$$

(b)

$$\boxed{\left.\frac{\partial z}{\partial y}\right|_{(1,1,1)} = -\frac{5}{3}}.$$

D. Multivariable Chain Rule

(a)

$$\boxed{F_x(1,-1) = 8}, \qquad \boxed{F_y(1,-1) = -1}.$$

(b)

$$\boxed{g_t(1,2) = 6}.$$

(c)

$$\boxed{f_r = 2rz\sin\theta\cos\theta + 2re^{r^2}},$$

$$\boxed{f_\theta = r^2 z(\cos^2\theta - \sin^2\theta)}, \qquad \boxed{f_z = r^2\sin\theta\cos\theta}.$$